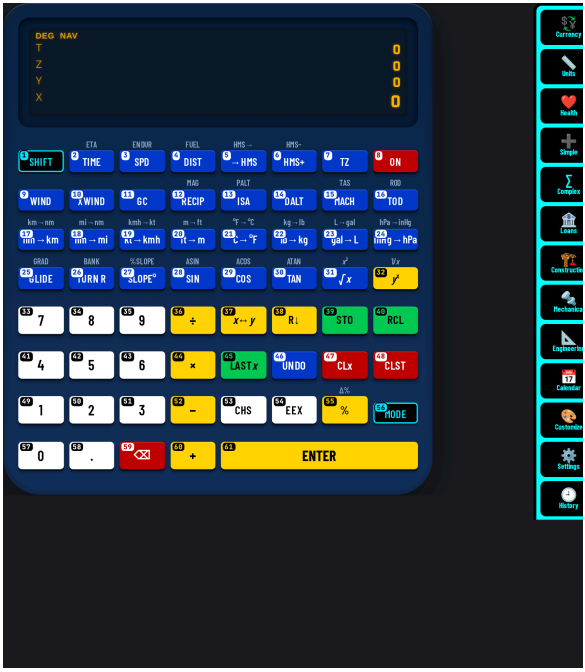
U-45 CALCULATORS

User Manual

Calculator #37 Navigation & Flight Computer RPN (aviation)

Scientific – RPN



Complete kit: functions · numbered key map · worked examples · tricks & hacks · shortcuts · common mistakes · practice drills

About this manual

This manual belongs to the **U-45 CALCULATORS** app and describes calculator **#37 — Navigation & Flight Computer RPN (aviation)** exactly as it behaves in the app. Every worked example and every drill answer in this book was run on the app's own keys by an automated checker before printing; only results that matched were kept.

How key presses are written

Keys are written as a chip showing the **key number** and its face, for example [#1 SHIFT]. Numbers follow the numbering printed on the keys (switch it on with "# On"). A shifted function is written as two chips: the shift key, then the key that carries the function.

Entry style of this model

four-level RPN navigation computer: enter the inputs bottom-up (Y then X; for 4-input tools T, Z, Y, X) and press the tool key.

Chapters

1. Getting started
2. Features
3. Feature guide (2b)
4. Numbered key map
5. Key-by-key reference
6. Worked examples
7. Tricks & hacks
8. Keyboard shortcuts
9. Common mistakes
10. Practice drills

11. Appendix A — Drill answers
12. Appendix B — Function finder (A–Z)
13. Appendix C — Glossary
14. Appendix D — Model notes & limits
15. Appendix E — Quick-reference card

1 · Getting started

Open U-45 CALCULATORS, choose **#37** from the home screen (or the jump list at the top), and the calculator appears with its numbered, colour-coded keys. Click or tap keys, or type on a keyboard.

Numbers on every key

Every key carries a small number in the same reading order as this manual (left → right, top → bottom; the navigation pad counts ▲ ▼ ◀ ▶ then its centre). Use "# On / # Off" in the bar above the calculator to show or hide them — the setting is remembered.

High-contrast colours

Keys are coloured by job: white = digits, yellow = operators/execute, blue = functions, green = memory/variables, red = clear/delete/power, black with cyan = shift/modes/menus/navigation. Press "Classic" to return to the original model colours and "Hi-Contrast" to come back.

Three typefaces

Display text uses the U45 Display face, key legends use U45 Keys (a condensed face that stays readable on small keys) and key numbers / helper text use U45 Mono. All three are bundled inside the app, so they work offline.

Copy the display

The Copy button copies exactly what the display shows (indicators that are switched off are skipped) so you can paste a result into a message or spreadsheet.

Fit or 1:1

Fit scales the calculator to your screen; 1:1 shows it at its design size. On a phone, Fit is the comfortable choice.

Reset

Reset rebuilds this calculator from scratch: display, memory, stack and modes return to their start-up state. Use it before each worked example in this manual.

Keyboard

On a computer you can type: digits, + − * / ^ ( ), Enter or = to execute, Backspace to delete, Esc to clear, arrows to move. Letters reach common functions (see the Keyboard shortcuts chapter).

Tooltips

Hover over (or long-press) a key to see "#number name".

Tools rail (v1.4)

A cyan icon rail on the right of every calculator opens 15 panels: Currency (166 currencies, rates auto-updated from open.er-api.com or Frankfurter when online and cached with a date; the bundled dated snapshot is used offline), Units (exact SI factors), Health (published formulas, not medical advice), Simple, Complex, Loans, Construction, Mechanical and Engineering (30+ numbered tools each), Calendar generator, Customize (26 options, saved per calculator), Settings (31 options), History, Hide and View. On narrow phones the rail shows icons only.

History (v1.4)

Every calculation is logged per calculator with a date and time. Open History on the rail to search, re-use (types the input back into the calculator), copy, delete one, select several and delete, or delete all (asks for confirmation). History is stored encrypted on this device (AES-GCM); exporting it needs the passcode.

Camera Solve (v1.4)

The green CS button between Restore and Social reads a problem from the camera, a photo, a PDF or a text file. Recognition (Tesseract OCR, English) runs offline on the device; an online language download is only used if you switch it on in Settings. Check and edit the recognised text, then Solve shows the steps offline. OCR is not available when the app is opened as a plain file:// page in a browser.

Layouts and view modes (v1.4)

The layout selector between Fit and # offers Auto plus 12 layouts: 3D classic, 3D high-contrast, Classic, Pocket, Compact, Expanded, Desktop wide, Split, One-hand left, One-hand right, Large print and Minimal. Auto picks one for your screen and your choice is remembered per calculator. The View panel hides the top bars, the tool bar, the rail or everything (full screen); the round yellow ⤢ button at bottom-left (or Esc) always restores the normal view.

Hide items (v1.4)

The Hide panel lists every button, rail icon and section of this calculator as tick boxes. Hiding is instant; unhiding asks for the app passcode.

Image protection (v1.4)

Right-click, drag and long-press saving of images is blocked inside the app. Saving or sharing an image export (PNG, JPG, QR, calendar), printing a calendar, and downloading or sharing the app itself ask for the passcode, and exported images carry a U-45 watermark. Honest note: this cannot stop a screenshot or screen recording made by the operating system or another device.

Works offline

U-45 CALCULATORS is an offline app: after the first load everything — including fonts — comes from the device.

Colour groups on this calculator

Colour	Group	Keys on this model
num	Digits & number entry	#33, #34, #35, #41, #42, #43, #49, #50, #51, #53, #54, #57, #58
op	Operators & execute	#32, #36, #37, #38, #44, #52, #55, #60, #61
fn	Functions	#2, #3, #4, #5, #6, #7, #9, #10, #11, #12, #13, #14, #15, #16, #17, #18, #19, #20, #21, #22, #23, #24, #25, #26, #27, #28, #29, #30, #31, #46
mem	Memory, variables & registers	#39, #40, #45

clr	**Clear, delete & power**	**#8, #47, #48, #59**
mod	**Shift, modes, menus & navigation**	**#1, #56**

2 · Features

48 features: 34 live on the keys, 9 app/UI features, 5 planned (documented but not yet working — they are clearly marked and never used in examples).

1. LIVE **Four-level RPN stack X Y Z T with ENTER, x ↔ y, R↓, LAST X and UNDO**
2. LIVE **Decimal hours → H.MMSS (→HMS) and back (HMS→)**
3. LIVE **HMS addition and subtraction**
4. LIVE **Time from distance and speed**
5. LIVE **Speed from distance and time**
6. LIVE **Distance from speed and time**
7. LIVE **Fuel used from burn rate and time**
8. LIVE **Endurance (H.MMSS) from fuel on board and burn rate**
9. LIVE **Great-circle distance between two lat/lon points**
10. LIVE **Initial great-circle course**
11. LIVE **Reciprocal heading (±180°)**
12. LIVE **True → magnetic with variation**
13. LIVE **Wind triangle: heading, ground speed and wind-correction angle**
14. LIVE **Headwind and crosswind components**
15. LIVE **Nautical miles ↔ kilometres**
16. LIVE **Nautical miles ↔ statute miles**
17. LIVE **Knots ↔ km/h**
18. LIVE **Feet ↔ metres**
19. LIVE **°C ↔ °F**
20. LIVE **Pounds ↔ kilograms**

21. LIVE US gallons ↔ litres
22. LIVE inHg ↔ hPa
23. LIVE ISA temperature at an altitude
24. LIVE Pressure altitude from field elevation and altimeter setting
25. LIVE Density altitude from pressure altitude and OAT
26. LIVE Top-of-descent distance (3 nm per 1 000 ft)
27. LIVE Rate of descent for a glide-path angle and ground speed
28. LIVE Glide distance from altitude and glide ratio
29. LIVE Climb gradient (ft/nm) from rate and ground speed
30. LIVE Mach number from TAS and OAT
31. LIVE TAS estimate from CAS and altitude (2 % per 1 000 ft)
32. LIVE Standard-rate turn radius and bank angle
33. LIVE Slope % ↔ degrees
34. LIVE Time-zone conversion and ETA in H.MMSS
35. APP/UI Every key numbered 1…N in reading order — matches the key map in the user manual
36. APP/UI High-contrast 6-colour key groups (digits, operators, functions, memory, clear, modes) with a “Classic” toggle
37. APP/UI Three bundled offline fonts: Oxanium (display/titles), Barlow Condensed (key faces), JetBrains Mono (key numbers & helpers)
38. APP/UI Show / hide key numbers with one button (remembered between sessions)
39. APP/UI Copy the current display text to the clipboard
40. APP/UI Physical-keyboard shortcuts: digits, + – * / ^ ( ) . Enter Backspace Esc arrows, s c t l n r ! % e p q i

41. APP/UI **Fit / 1:1 zoom that never pushes keys off a phone screen**
42. APP/UI **Reference photo viewer and tool-tip name on every key**
43. APP/UI **Reset = fresh power-on state; works fully offline (PWA cache / Android APK)**
44. PLANNED **Stack history view**
45. PLANNED **Undo**
46. PLANNED **Equation library**
47. PLANNED **Speech read-out**
48. PLANNED **Colour-blind key theme**

2b · Feature guide

Calculator #37 is a four-level RPN navigation computer: enter the inputs bottom-up (Y then X; for 4-input tools T, Z, Y, X) and press the tool key. Each card below is one of its 34 working features, with the keys that reach it and, where one exists, a verified example.

F1 Four-level RPN stack X Y Z T with ENTER, x ↔ y, R↓, LAST X and UNDO

Where: #61 ENTER ENTER — terminates digit entry and pushes X up the stack (RPN) / confirms a value; #32 yx y^x — raise Y to the power X (RPN: base first, then exponent); #38 R↓ R↓ — roll the stack down

See it in action: Flight time — *120 nm ENTER 100 kt TIME* = 1.2; ISA sea-level temperature in °F — *15 °C → °F* = 59

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F2 Decimal hours → H.MMSS (→ HMS) and back (HMS →)

Where: #5 →HMS Decimal hours to H.MMSS; #6 HMS+ hmsadd; #2 TIME Time = Y dist ÷ X speed

See it in action: Decimal hours to H.MMSS — *1.5* → *HMS* = 1.3

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F3 HMS addition and subtraction

Where: #5 →HMS Decimal hours to H.MMSS; #6 HMS+ hmsadd; #52 − Subtraction −

See it in action: Decimal hours to H.MMSS — *1.5* → *HMS* = 1.3

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F4 Time from distance and speed

Where: #4 DIST Distance = Y speed × X time; #2 TIME Time = Y dist ÷ X speed; #3 SPD Speed = Y dist ÷ X time

See it in action: Flight time — *120 nm ENTER 100 kt TIME* = 1.2; Great-circle distance along the equator — *0 ENTER 0 ENTER 0 ENTER 1 GC* = 60.04046073

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F5 Speed from distance and time

Where: #4 DIST Distance = Y speed × X time; #2 TIME Time = Y dist ÷ X speed; #3 SPD Speed = Y dist ÷ X time

See it in action: Flight time — *120 nm ENTER 100 kt TIME* = 1.2; Great-circle distance along the equator — *0 ENTER 0 ENTER 0 ENTER 1 GC* = 60.04046073

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F6 Distance from speed and time

Where: #4 DIST Distance = Y speed × X time; #2 TIME Time = Y dist ÷ X speed; #3 SPD Speed = Y dist ÷ X time

See it in action: Flight time — *120 nm ENTER 100 kt TIME* = 1.2; Great-circle distance along the equator — *0 ENTER 0 ENTER 0 ENTER 1 GC* = 60.04046073

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F7 Fuel used from burn rate and time

Where: #3 SPD Speed = Y dist ÷ X time; #4 DIST Distance = Y speed × X time; #2 TIME Time = Y dist ÷ X speed

See it in action: Flight time — *120 nm ENTER 100 kt TIME* = 1.2

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F8 Endurance (H.MMSS) from fuel on board and burn rate

Where: #3 SPD Speed = Y dist ÷ X time; #4 DIST Distance = Y speed × X time; #2 TIME Time = Y dist ÷ X speed

See it in action: Decimal hours to H.MMSS — *1.5 → HMS* = 1.3

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F9 Great-circle distance between two lat/lon points

Where: #11 GC Great circle: T lat1, Z lon1, Y lat2, X lon2; #4 DIST Distance = Y speed × X time; #16 TOD Descent distance for X ft

See it in action: Great-circle distance along the equator — *0 ENTER 0 ENTER 0 ENTER 1 GC* = 60.04046073; Top-of-descent distance — *9000 ft TOD* = 27

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F10 Initial great-circle course

Where: #11 GC **Great circle: T lat1, Z lon1, Y lat2, X lon2**

See it in action: Great-circle distance along the equator — *0 ENTER 0 ENTER 0 ENTER 1 GC* = 60.04046073

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F11 Reciprocal heading (±180°)

Where: #32 yx **y^x — raise Y to the power X (RPN: base first, then exponent)**

See it in action: Reciprocal heading — *90 RECIP* = 270

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F12 True → magnetic with variation

Where: #12 RECIP **recip**

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F13 Wind triangle: heading, ground speed and wind-correction angle

Where: #10 XWIND Head/crosswind: Y angle, X wind speed; #9 WIND Wind triangle form; #2 TIME Time = Y dist ÷ X speed

See it in action: Reciprocal heading — *90 RECIP* = 270

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F14 Headwind and crosswind components

Where: #10 XWIND Head/crosswind: Y angle, X wind speed

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F15 Nautical miles ↔ kilometres

Where: from the TOOLS / menu screens of this model (see chapter 4 for the menu key)

See it in action: Nautical miles to km — *10 nm → km* = 18.52

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F16 Nautical miles ↔ statute miles

Where: from the TOOLS / menu screens of this model (see chapter 4 for the menu key)

See it in action: Nautical miles to km — *10 nm → km* = 18.52

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F17 Knots ↔ km/h

Where: from the TOOLS / menu screens of this model (see chapter 4 for the menu key)

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F18 Feet ↔ metres

Where: from the TOOLS / menu screens of this model (see chapter 4 for the menu key)

See it in action: Feet to metres — *1000 ft → m* = 304.8

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F19 °C ↔ °F

Where: from the TOOLS / menu screens of this model (see chapter 4 for the menu key)

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F20 Pounds ↔ kilograms

Where: from the TOOLS / menu screens of this model (see chapter 4 for the menu key)

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F21 US gallons ↔ litres

Where: from the TOOLS / menu screens of this model (see chapter 4 for the menu key)

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F22 inHg ↔ hPa

Where: #24 inHg → hPa inhg2hpa; #13 ISA ISA temperature at X ft

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F23 ISA temperature at an altitude

Where: #13 ISA ISA temperature at X ft; #15 MACH Mach: Y TAS kt, X OAT °C

See it in action: ISA sea-level temperature in °F — *15 °C → °F* = 59

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F24 Pressure altitude from field elevation and altimeter setting

Where: #13 ISA ISA temperature at X ft; #14 DALT Density alt: Y pressure alt, X OAT °C; #15 MACH Mach: Y TAS kt, X OAT °C

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F25 Density altitude from pressure altitude and OAT

Where: #14 DALT Density alt: Y pressure alt, X OAT °C; #15 MACH Mach: Y TAS kt, X OAT °C; #13 ISA ISA temperature at X ft

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F26 Top-of-descent distance (3 nm per 1 000 ft)

Where: #16 TOD Descent distance for X ft; #4 DIST Distance = Y speed × X time; #25 GLIDE Glide distance: Y alt ft, X ratio

See it in action: Great-circle distance along the equator — *0 ENTER 0 ENTER 0 ENTER 1 GC* = 60.04046073; Top-of-descent distance — *9000 ft TOD* = 27

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F27 Rate of descent for a glide-path angle and ground speed

Where: #16 TOD Descent distance for X ft; #3 SPD Speed = Y dist ÷ X time; #4 DIST Distance = Y speed × X time

See it in action: Top-of-descent distance — *9000 ft TOD* = 27

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F28 Glide distance from altitude and glide ratio

Where: #25 GLIDE Glide distance: Y alt ft, X ratio; #4 DIST Distance = Y speed × X time; #15 MACH Mach: Y TAS kt, X OAT °C

See it in action: Great-circle distance along the equator — *0 ENTER 0 ENTER 0 ENTER 1 GC* = 60.04046073; Top-of-descent distance — *9000 ft TOD* = 27

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F29 Climb gradient (ft/nm) from rate and ground speed

Where: #3 SPD Speed = Y dist ÷ X time; #4 DIST Distance = Y speed × X time; #25 GLIDE Glide distance: Y alt ft, X ratio

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F30 Mach number from TAS and OAT

Where: #15 MACH Mach: Y TAS kt, X OAT °C; #14 DALT Density alt: Y pressure alt, X OAT °C; #53 CHS Change sign (+/−) of the number being typed or the current result

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F31 TAS estimate from CAS and altitude (2 % per 1 000 ft)

Where: #15 MACH Mach: Y TAS kt, X OAT °C

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F32 Standard-rate turn radius and bank angle

Where: #26 TURN R Standard-rate turn radius (nm) at X kt; #16 TOD Descent distance for X ft; #3 SPD Speed = Y dist ÷ X time

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F33 Slope % ↔ degrees

Where: #27 SLOPE° Slope % to degrees

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F34 Time-zone conversion and ETA in H.MMSS

Where: #2 TIME Time = Y dist ÷ X speed; #3 SPD Speed = Y dist ÷ X time; #4 DIST Distance = Y speed × X time

See it in action: Flight time — *120 nm ENTER 100 kt TIME* = 1.2; Decimal hours to H.MMSS — *1.5 → HMS* = 1.3

Try it: Press Reset, use the key(s) above, and write the display here:

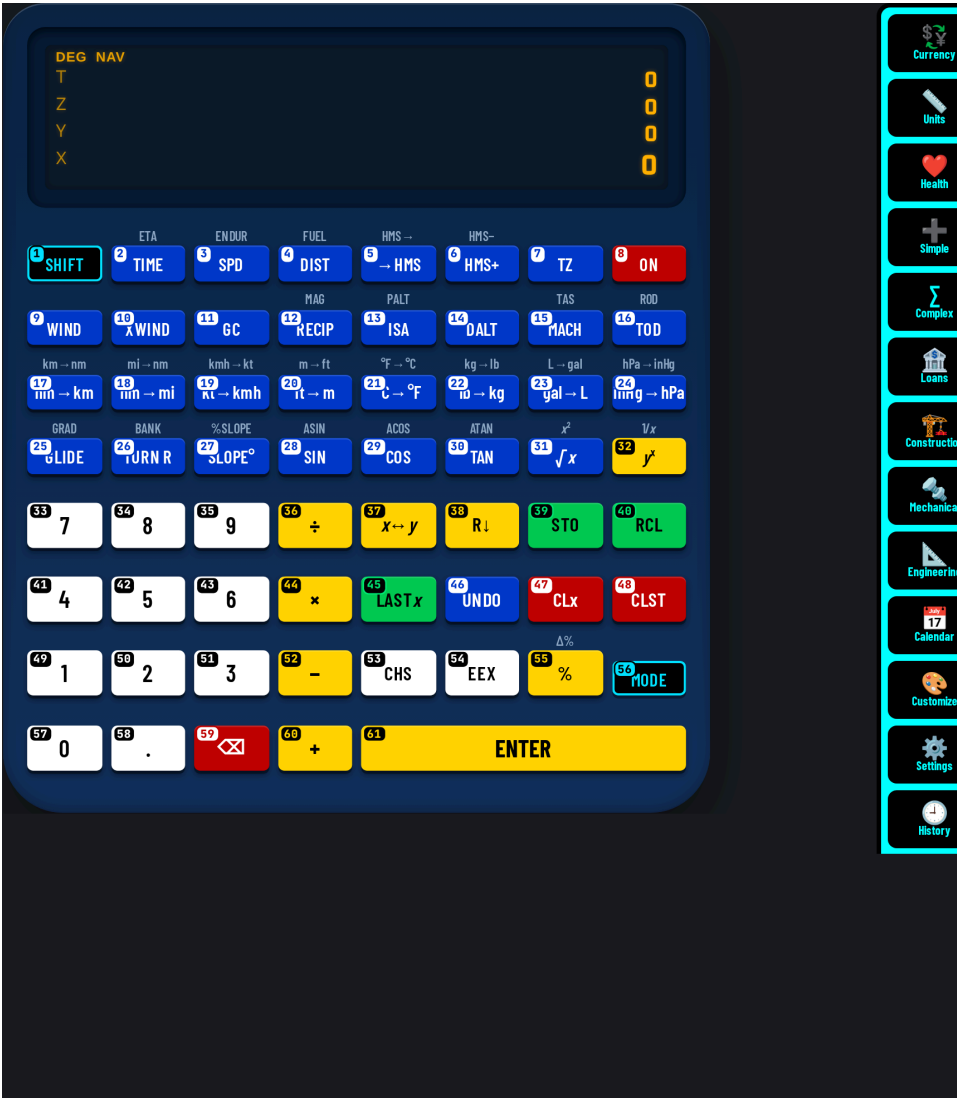
_____

Your notes:

_____

_____

3 · Numbered key map



Calculator #37 with key numbers shown (Hi-Contrast colours).

Key table

"Shift" = 2nd / f / SHIFT layer; "Alpha" = ALPHA layer; "3rd" = 3rd / g layer. A dot means the layer is empty on that key.

#	Face	Shift	Alpha	3rd	Group
1	SHIFT	·	·	·	mod
2	TIME	ETA	·	·	fn
3	SPD	ENDUR	·	·	fn
4	DIST	FUEL	·	·	fn
5	→ HMS	HMS →	·	·	fn
6	HMS+	HMS−	·	·	fn
7	TZ	·	·	·	fn
8	ON	·	·	·	clr
9	WIND	·	·	·	fn
10	XWIND	·	·	·	fn
11	GC	·	·	·	fn
12	RECIP	MAG	·	·	fn
13	ISA	PALT	·	·	fn
14	DALT	·	·	·	fn
15	MACH	TAS	·	·	fn
16	TOD	ROD	·	·	fn
17	nm → km	km → nm	·	·	fn
18	nm → mi	mi → nm	·	·	fn
19	kt → kmh	kmh → kt	·	·	fn
20	ft → m	m → ft	·	·	fn
21	°C → °F	°F → °C	·	·	fn
22	lb → kg	kg → lb	·	·	fn
23	gal → L	L → gal	·	·	fn

#	Face	Shift	Alpha	3rd	Group
24	inHg → hPa	hPa → inHg	·	·	fn
25	GLIDE	GRAD	·	·	fn
26	TURN R	BANK	·	·	fn
27	SLOPE°	%SLOPE	·	·	fn
28	SIN	ASIN	·	·	fn
29	COS	ACOS	·	·	fn
30	TAN	ATAN	·	·	fn
31	√x	x^2	·	·	fn
32	yx	1/x	·	·	op
33	7	·	·	·	num
34	8	·	·	·	num
35	9	·	·	·	num
36	÷	·	·	·	op
37	x ↔ y	·	·	·	op
38	R↓	·	·	·	op
39	STO	·	·	·	mem
40	RCL	·	·	·	mem
41	4	·	·	·	num
42	5	·	·	·	num
43	6	·	·	·	num
44	×	·	·	·	op
45	LASTx	·	·	·	mem
46	UNDO	·	·	·	fn
47	CLx	·	·	·	clr
48	CLST	·	·	·	clr
49	1	·	·	·	num
50	2	·	·	·	num
51	3	·	·	·	num

#	Face	Shift	Alpha	3rd	Group
52	−	·	·	·	op
53	CHS	·	·	·	num
54	EEX	·	·	·	num
55	%	Δ%	·	·	op
56	MODE	·	·	·	mod
57	0	·	·	·	num
58	.	·	·	·	num
59	⌫	·	·	·	clr
60	+	·	·	·	op
61	ENTER	·	·	·	op

4 · Key-by-key reference

Every key on calculator #37, in number order, with every function it carries in this app.

#1 SHIFT `Shift, modes, menus & navigation`

Primary: SHIFT / 2nd — the next key uses its upper (shift-coloured) function

#2 TIME `Functions`

Primary: Time = Y dist ÷ X speed

shifted (2nd / f / SHIFT layer) (press #1 first): ETA — ETA = Y dep + X ete (H.MMSS)

#3 SPD `Functions`

Primary: Speed = Y dist ÷ X time

shifted (2nd / f / SHIFT layer) (press #1 first): ENDUR — Endurance = Y fuel ÷ X rate

#4 DIST `Functions`

Primary: Distance = Y speed × X time

shifted (2nd / f / SHIFT layer) (press #1 first): FUEL — Fuel = Y rate × X time

#5 → HMS `Functions`

Primary: Decimal hours to H.MMSS

shifted (2nd / f / SHIFT layer) (press #1 first): HMS → — H.MMSS to decimal hours

#6 HMS+ Functions

Primary: hmsadd

shifted (2nd / f / SHIFT layer) (press #1 first): HMS− — Subtract H.MMSS

#7 TZ Functions

Primary: UTC (Y) + offset (X)

#8 ON Clear, delete & power

Primary: ON — power on

#9 WIND Functions

Primary: Wind triangle form

#10 XWIND Functions

Primary: Head/crosswind: Y angle, X wind speed

#11 GC Functions

Primary: Great circle: T lat1, Z lon1, Y lat2, X lon2

#12 RECIP Functions

Primary: recip

shifted (2nd / f / SHIFT layer) (press #1 first): MAG — Magnetic = Y true − X variation (E+)

#13 ISA Functions

Primary: ISA temperature at X ft

shifted (2nd / f / SHIFT layer) (press #1 first): PALT — Pressure alt: Y elevation, X altimeter inHg

#14 DALT Functions

Primary: Density alt: Y pressure alt, X OAT °C

#15 MACH Functions

Primary: Mach: Y TAS kt, X OAT °C

shifted (2nd / f / SHIFT layer) (press #1 first): TAS — TAS: Y CAS, X altitude ft

#16 TOD Functions

Primary: Descent distance for X ft

shifted (2nd / f / SHIFT layer) (press #1 first): ROD — Rate of descent: Y GS, X angle

#17 nm → km Functions

Primary: nm2km

shifted (2nd / f / SHIFT layer) (press #1 first): km → nm — km2nm

#18 nm → mi Functions

Primary: nm2mi

shifted (2nd / f / SHIFT layer) (press #1 first): mi → nm — mi2nm

#19 kt → kmh Functions

Primary: kt2kmh

shifted (2nd / f / SHIFT layer) (press #1 first): kmh → kt — kmh2kt

#20 ft → m Functions

Primary: ft2m

shifted (2nd / f / SHIFT layer) (press #1 first): m → ft — m2ft

#21 °C → °F Functions

Primary: °C → °F

shifted (2nd / f / SHIFT layer) (press #1 first): °F → °C — °F → °C

#22 lb → kg Functions

Primary: lb2kg

shifted (2nd / f / SHIFT layer) (press #1 first): kg → lb — kg2lb

#23 gal → L Functions

Primary: gal2l

shifted (2nd / f / SHIFT layer) (press #1 first): L → gal — l2gal

#24 inHg → hPa Functions

Primary: inhg2hpa

shifted (2nd / f / SHIFT layer) (press #1 first): hPa → inHg — hpa2inhg

#25 GLIDE Functions

Primary: Glide distance: Y alt ft, X ratio

shifted (2nd / f / SHIFT layer) (press #1 first): GRAD — Climb gradient: Y fpm, X GS

#26 TURN R Functions

Primary: Standard-rate turn radius (nm) at X kt

shifted (2nd / f / SHIFT layer) (press #1 first): BANK — Standard-rate bank angle

#27 SLOPE° Functions

Primary: Slope % to degrees

shifted (2nd / f / SHIFT layer) (press #1 first): %SLOPE — Degrees to slope %

#28 SIN Functions

Primary: Sine (uses the current angle unit)

shifted (2nd / f / SHIFT layer) (press #1 first): ASIN — Inverse sine $\sin^{-1}$

Keyboard: s

#29 COS Functions

Primary: Cosine (current angle unit)

shifted (2nd / f / SHIFT layer) (press #1 first): ACOS — Inverse cosine $\cos^{-1}$

Keyboard: c

#30 TAN Functions

Primary: Tangent (current angle unit)

shifted (2nd / f / SHIFT layer) (press #1 first): ATAN — Inverse tangent $\tan^{-1}$

Keyboard: t

#31 √x Functions

Primary: Square root √

shifted (2nd / f / SHIFT layer) (press #1 first): x^2 — Square x^2

Keyboard: r

#32 yx Operators & execute

Primary: y^x — raise Y to the power X (RPN: base first, then exponent)

shifted (2nd / f / SHIFT layer) (press #1 first): 1/x — Reciprocal x^{-1} (1/x)

Keyboard: ^

#33 7 Digits & number entry

Primary: Digit 7

Keyboard: 7

#34 8 Digits & number entry

Primary: Digit 8

Keyboard: 8

#35 9 Digits & number entry

Primary: Digit 9

Keyboard: 9

#36 ÷ Operators & execute

Primary: Division ÷

Keyboard: /

#37 x ↔ y Operators & execute

Primary: x≷y — exchange X and Y

#38 R↓ Operators & execute

Primary: R↓ — roll the stack down

#39 STO Memory, variables & registers

Primary: STO — store the value into a memory / variable (press the target next)

#40 RCL Memory, variables & registers

Primary: RCL — recall a memory / variable

#41 4 Digits & number entry

Primary: Digit 4

Keyboard: 4

#42 5 Digits & number entry

Primary: Digit 5

Keyboard: 5

#43 6 Digits & number entry

Primary: Digit 6

Keyboard: 6

#44 × Operators & execute

Primary: Multiplication ×

Keyboard: * x

#45 LASTx Memory, variables & registers

Primary: LASTx — recall the X value used by the last operation

#46 UNDO Functions

Primary: undo

#47 CLx Clear, delete & power

Primary: CLx — clear X

#48 CLST Clear, delete & power

Primary: AC — clear all of the current calculation

Keyboard: Escape

#49 1 Digits & number entry

Primary: Digit 1

Keyboard: 1

#50 **2** Digits & number entry

Primary: Digit 2

Keyboard: 2

#51 **3** Digits & number entry

Primary: Digit 3

Keyboard: 3

#52 **–** Operators & execute

Primary: Subtraction –

Keyboard: -

#53 **CHS** Digits & number entry

Primary: Change sign (+/−) of the number being typed or the current result

#54 **EEX** Digits & number entry

Primary: EEX — enter a power-of-ten exponent for the number in X

Keyboard: e

#55 **%** Operators & execute

Primary: Percent %

shifted (2nd / f / SHIFT layer) (press #1 first): Δ% — Δ% — percent change

Keyboard: %

#56 MODE `Shift, modes, menus & navigation`

Primary: SETUP — angle unit, number format, display options

#57 0 `Digits & number entry`

Primary: Digit 0

Keyboard: `0`

#58 . `Digits & number entry`

Primary: Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored

Keyboard: `.` `,`

#59 ⌫ `Clear, delete & power`

Primary: Backspace — delete the last digit typed

Keyboard: `Backspace` `Delete`

#60 + `Operators & execute`

Primary: Addition +

Keyboard: `+`

#61 ENTER Operators & execute

Primary: ENTER — terminates digit entry and pushes X up the stack (RPN) / confirms a value

Keyboard: Enter

5 · Worked examples

8 examples, each run on this calculator's keys by the U-45 checker. Follow the chips in order.

Example 5.1 — Flight time

Calculate **120 nm ENTER 100 kt TIME**. Press **Reset** first.

`#49` 1 → `#50` 2 → `#57` 0 → `#61` ENTER → `#49` 1 → `#57` 0 → `#57` 0 → `#2` TIME

Display shows `X1.2` — expected 1.2. ✔ verified on the app.

Example 5.2 — Reciprocal heading

Calculate **90 RECIP**. Press **Reset** first.

`#35` 9 → `#57` 0 → `#12` RECIP

Display shows `X270` — expected 270. ✔ verified on the app.

Example 5.3 — Nautical miles to km

Calculate **10 nm → km**. Press **Reset** first.

`#49` 1 → `#57` 0 → `#17` nm → km

Display shows `X18.52` — expected 18.52. ✔ verified on the app.

Example 5.4 — Decimal hours to H.MMSS

Calculate **1.5 →HMS**. Press **Reset** first.

#49 1 → **#58** . → **#42** 5 → **#5** →HMS

Display shows `X1.3` — expected 1.3. ✔ verified on the app.

Example 5.5 — Feet to metres

Calculate **1000 ft→m**. Press **Reset** first.

#49 1 → **#57** 0 → **#57** 0 → **#57** 0 → **#20** ft→m

Display shows `X304.8` — expected 304.8. ✔ verified on the app.

Example 5.6 — ISA sea-level temperature in °F

Calculate **15 °C→°F**. Press **Reset** first.

#49 1 → **#42** 5 → **#21** °C→°F

Display shows `X59` — expected 59. ✔ verified on the app.

Example 5.7 — Great-circle distance along the equator

Calculate **0 ENTER 0 ENTER 0 ENTER 1 GC**. Press **Reset** first.

#57 0 → **#61** ENTER → **#57** 0 → **#61** ENTER → **#57** 0 → **#61** ENTER → **#49** 1 → **#11** GC

Display shows `X60.04046073` — expected 60.04046073. ✔ verified on the app.

Example 5.8 — Top-of-descent distance

Calculate 9000 ft TOD. Press Reset first.

#35 9 → #57 0 → #57 0 → #57 0 → #16 TOD

Display shows X27 **— expected 27. ✔ verified on the app.**

6 · Tricks & hacks

Hack 1: No equals key

RPN works number, number, operator: 3 ENTER 4 + = 7. Results stay on the stack for the next step.

Hack 2: Chain without ENTER

After an operation, just type the next number and operator: 3 ENTER 4 + 5 × gives (3+4)×5 = 35.

Hack 3: Swap to fix order

If you entered numbers in the wrong order for − ÷ or y^x, press x≷y before the operator.

Hack 4: LASTx to reuse

LASTx brings back the X value of the previous operation — useful for a × a.

Hack 5: Stack as memory

Up to four values are kept; R↓ rolls through them.

Hack 6: Constant via stack top

On 4-level stacks the T register copies down, so a constant multiplier can be reused by pressing × repeatedly.

Hack 7: Hidden functions on this model

26 extra functions sit on shifted layers. Press the layer key once, then the key: for example #1 then #2 → ETA; #1 then #3 → ENDUR; #1 then #4 → FUEL; #1 then #5 → HMS→.

Hack 8: Find any key by number

Use Appendix B (function finder): look up the function, read its key number, then find that number on the keypad. Numbers run left→right, top→bottom from #1 to #61.

Hack 9: Type faster on a computer

25 keys on this model have a keyboard shortcut (see chapter 7). Typing digits and operators is much faster than clicking.

Hack 10: Check before you trust

Estimate first (round each number) and compare with the display. If the answer is 10× or 100× off, a decimal point or percent key was missed.

Hack 11: Copy results out

Press Copy in the bar above the calculator to put the display text on the clipboard instead of retyping it.

Hack 12: Colour blind or low vision?

Hi-Contrast colours use black/white/yellow/blue/green/red with black or white text for maximum contrast; key numbers stay on in both colour schemes.

7 · Keyboard shortcuts

With the calculator open on a computer, these physical keys press the numbered keys below (Ctrl/Alt/⌘ combinations are left to the browser). Shifted functions are reached by clicking the shift key first.

Keyboard	#	Key	Does
s	#28	SIN	Sine (uses the current angle unit)
c	#29	COS	Cosine (current angle unit)
t	#30	TAN	Tangent (current angle unit)
r	#31	√x	Square root √
^	#32	yx	y^x — raise Y to the power X (RPN: base first, then exponent)
7	#33	7	Digit 7
8	#34	8	Digit 8
9	#35	9	Digit 9
/	#36	÷	Division ÷
4	#41	4	Digit 4
5	#42	5	Digit 5
6	#43	6	Digit 6
* x	#44	×	Multiplication ×
Escape	#48	CLST	AC — clear all of the current calculation
1	#49	1	Digit 1
2	#50	2	Digit 2
3	#51	3	Digit 3
-	#52	−	Subtraction −
e	#54	EEX	EEX — enter a power-of-ten exponent for the number in X
%	#55	%	Percent %
0	#57	0	Digit 0

. ,	#58	.	Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored
Backspace Delete	#59	⌫	Backspace — delete the last digit typed
+	#60	+	Addition +
Enter	#61	ENTER	ENTER — terminates digit entry and pushes X up the stack (RPN) / confirms a value

App shortcuts

“# On / # Off”	show / hide key numbers
“Classic / Hi-Contrast”	switch key colours
“Copy”	copy display text
“Fit / 1:1”	scale to screen / design size
“Reset”	restart this calculator
Jump list	switch to another of the 45 calculators

8 · Common mistakes

Mistake 1: Pressing ENTER after an operation

ENTER after + duplicates the result. Usually you want to type the next number directly.

Mistake 2: Wrong order for – and ÷

Y – X and Y ÷ X: enter the first operand first. 10 ENTER 4 – = 6.

Mistake 3: CHS vs –

Make a number negative with CHS/+/−, not with −.

Mistake 4: Prefix left active

If f / g / SHIFT is shown, the next key is shifted. Clear it with the prefix-clear function.

Mistake 5: Forgetting Reset between problems

Old values (memory, stack, open brackets, worksheet values) can leak into the next calculation. Press Reset or clear before starting a new problem.

Mistake 6: Pressing the wrong layer

A shift / 2nd indicator that is still lit will turn the next key into its shifted function. Look at the display indicators before pressing.

9 · Practice drills

Set A — Calculation drills (8)

Work each one on calculator #37, then check your keys and the answer in Appendix A. Every answer was produced by this calculator.

1. 120 nm ENTER 100 kt TIME = _____
2. 90 RECIP = _____
3. 10 nm → km = _____
4. 1.5 → HMS = _____
5. 1000 ft → m = _____
6. 15 °C → °F = _____
7. 0 ENTER 0 ENTER 0 ENTER 1 GC = _____
8. 9000 ft TOD = _____

Set B — Key-finding drills (77)

Without looking at the table, find each key on the picture. Answers in Appendix A.

1. Which key is "SHIFT" (SHIFT / 2nd)? _____
2. Which key is "TIME" (Time = Y dist ÷ X speed)? _____
3. Which key is "SPD" (Speed = Y dist ÷ X time)? _____
4. Which key is "DIST" (Distance = Y speed × X time)? _____
5. Which key is " → HMS" (Decimal hours to H.MMSS)? _____
6. Which key is "HMS+" (hmsadd)? _____

7. Which key is “TZ” (UTC)? _____

8. Which key is “ON” (ON)? _____

9. Which key is “WIND” (Wind triangle form)? _____

10. Which key is “XWIND” (Head/crosswind: Y angle, X wind speed)? _____

11. Which key is “GC” (Great circle: T lat1, Z lon1, Y lat2, X lon2)? _____

12. Which key is “RECIP” (recip)? _____

13. Which key is “ISA” (ISA temperature at X ft)? _____

14. Which key is “DALT” (Density alt: Y pressure alt, X OAT °C)? _____

15. Which key is “MACH” (Mach: Y TAS kt, X OAT °C)? _____

16. Which key is “TOD” (Descent distance for X ft)? _____

17. Which key is “nm → km” (nm2km)? _____

18. Which key is “nm → mi” (nm2mi)? _____

19. Which key is “kt → kmh” (kt2kmh)? _____

20. Which key is “ft → m” (ft2m)? _____

21. Which key is “°C → °F” (°C → °F)? _____

22. Which key is “lb → kg” (lb2kg)? _____

23. Which key is “gal → L” (gal2l)? _____

24. Which key is “inHg → hPa” (inhg2hpa)? _____

25. Which key is “GLIDE” (Glide distance: Y alt ft, X ratio)? _____

26. Which key is “TURN R” (Standard-rate turn radius)? _____

27. Which key is "SLOPE°" (Slope % to degrees)? _____

28. Which key is "SIN" (Sine)? _____

29. Which key is "COS" (Cosine)? _____

30. Which key is "TAN" (Tangent)? _____

31. Which key is "√x" (Square root √)? _____

32. Which key is "yx" (y^x)? _____

33. Which key is "÷" (Division ÷)? _____

34. Which key is "x↔y" (x≷y)? _____

35. Which key is "R↓" (R↓)? _____

36. Which key is "STO" (STO)? _____

37. Which key is "RCL" (RCL)? _____

38. Which key is "×" (Multiplication ×)? _____

39. Which key is "LASTx" (LASTx)? _____

40. Which key is "UNDO" (undo)? _____

41. Which key is "CLx" (CLx)? _____

42. Which key is "CLST" (AC)? _____

43. Which key is "–" (Subtraction –)? _____

44. Which key is "CHS" (Change sign)? _____

45. Which key is "EEX" (EEX)? _____

46. Which key is "%" (Percent %)? _____

47. Which key is “MODE” (SETUP)? _____

48. Which key is “.” (Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored)? _____

49. Which key is “⌫” (Backspace)? _____

50. Which key is “+” (Addition +)? _____

51. Which key is “ENTER” (ENTER)? _____

52. How do you reach “ETA”? _____

53. How do you reach “ENDUR”? _____

54. How do you reach “FUEL”? _____

55. How do you reach “HMS →”? _____

56. How do you reach “HMS–”? _____

57. How do you reach “MAG”? _____

58. How do you reach “PALT”? _____

59. How do you reach “TAS”? _____

60. How do you reach “ROD”? _____

61. How do you reach “km → nm”? _____

62. How do you reach “mi → nm”? _____

63. How do you reach “kmh → kt”? _____

64. How do you reach “m → ft”? _____

65. How do you reach “°F → °C”? _____

66. How do you reach “kg → lb”? _____

67. How do you reach “L → gal”? _____

68. How do you reach “hPa → inHg”? _____

69. How do you reach “GRAD”? _____

70. How do you reach “BANK”? _____

71. How do you reach “%SLOPE”? _____

72. How do you reach “ASIN”? _____

73. How do you reach “ACOS”? _____

74. How do you reach “ATAN”? _____

75. How do you reach “x^2”? _____

76. How do you reach “1/x”? _____

77. How do you reach “Δ%”? _____

Appendix A — Drill answers

Set A

1. **1.2** (display `X1.2`)
   `#49` 1 → `#50` 2 → `#57` 0 → `#61` ENTER → `#49` 1 → `#57` 0 → `#57` 0 → `#2` TIME

2. **270** (display `X270`)
   `#35` 9 → `#57` 0 → `#12` RECIP

3. **18.52** (display `X18.52`)
   `#49` 1 → `#57` 0 → `#17` nm → km

4. **1.3** (display `X1.3`)
   `#49` 1 → `#58` . → `#42` 5 → `#5` → HMS

5. **304.8** (display `X304.8`)
   `#49` 1 → `#57` 0 → `#57` 0 → `#57` 0 → `#20` ft → m

6. **59** (display `X59`)
   `#49` 1 → `#42` 5 → `#21` °C → °F

7. **60.04046073** (display `X60.04046073`)
   `#57` 0 → `#61` ENTER → `#57` 0 → `#61` ENTER → `#57` 0 → `#61` ENTER → `#49` 1 → `#11` GC

8. **27** (display `X27`)
   `#35` 9 → `#57` 0 → `#57` 0 → `#57` 0 → `#16` TOD

Set B

1. #1
2. #2
3. #3
4. #4
5. #5
6. #6
7. #7
8. #8
9. #9
10. #10
11. #11
12. #12
13. #13
14. #14
15. #15
16. #16
17. #17
18. #18
19. #19
20. #20
21. #21
22. #22
23. #23
24. #24
25. #25
26. #26
27. #27
28. #28
29. #29
30. #30
31. #31
32. #32
33. #36
34. #37
35. #38
36. #39
37. #40
38. #44
39. #45
40. #46
41. #47
42. #48
43. #52
44. #53
45. #54
46. #55
47. #56
48. #58
49. #59
50. #60
51. #61
52. #1 SHIFT then #2 TIME
53. #1 SHIFT then #3 SPD
54. #1 SHIFT then #4 DIST
55. #1 SHIFT then #5 →HMS
56. #1 SHIFT then #6 HMS+
57. #1 SHIFT then #12 RECIP
58. #1 SHIFT then #13 ISA
59. #1 SHIFT then #15 MACH
60. #1 SHIFT then #16 TOD
61. #1 SHIFT then #17 nm→km
62. #1 SHIFT then #18 nm→mi
63. #1 SHIFT then #19 kt→kmh
64. #1 SHIFT then #20 ft→m
65. #1 SHIFT then #21 °C→°F
66. #1 SHIFT then #22 lb→kg
67. #1 SHIFT then #23 gal→L
68. #1 SHIFT then #24 inHg→hPa
69. #1 SHIFT then #25 GLIDE
70. #1 SHIFT then #26 TURN R
71. #1 SHIFT then #27 SLOPE°
72. #1 SHIFT then #28 SIN
73. #1 SHIFT then #29 COS
74. #1 SHIFT then #30 TAN
75. #1 SHIFT then #31 √x
76. #1 SHIFT then #32 yx
77. #1 SHIFT then #55 %

Appendix B — Function finder (A–Z)

Function	Label	Keys
°C → °F	°C→°F	#21
°F → °C	°F→°C	#1 → #21
AC — clear all of the current calculation	CLST	#48
Addition +	+	#60
Backspace — delete the last digit typed	⌫	#59
Change sign (+/−) of the number being typed or the current result	CHS	#53
Climb gradient: Y fpm, X GS	GRAD	#1 → #25
CLx — clear X	CLx	#47
Cosine (current angle unit)	COS	#29
Decimal hours to H.MMSS	→HMS	#5
Decimal point. Pressing it first gives "0."; a second point in the same number is ignored	.	#58
Degrees to slope %	%SLOPE	#1 → #27
Density alt: Y pressure alt, X OAT °C	DALT	#14
Descent distance for X ft	TOD	#16
Digit 0	0	#57
Digit 1	1	#49
Digit 2	2	#50
Digit 3	3	#51
Digit 4	4	#41
Digit 5	5	#42
Digit 6	6	#43
Digit 7	7	#33
Digit 8	8	#34

Digit 9	9	#35
Distance = Y speed × X time	DIST	#4
Division ÷	÷	#36
EEX — enter a power-of-ten exponent for the number in X	EEX	#54
Endurance = Y fuel ÷ X rate	ENDUR	#1 → #3
ENTER — terminates digit entry and pushes X up the stack (RPN) / confirms a value	ENTER	#61
ETA = Y dep + X ete (H.MMSS)	ETA	#1 → #2
ft2m	ft → m	#20
Fuel = Y rate × X time	FUEL	#1 → #4
gal2l	gal → L	#23
Glide distance: Y alt ft, X ratio	GLIDE	#25
Great circle: T lat1, Z lon1, Y lat2, X lon2	GC	#11
H.MMSS to decimal hours	HMS →	#1 → #5
Head/crosswind: Y angle, X wind speed	XWIND	#10
hmsadd	HMS+	#6
hpa2inhg	hPa → inHg	#1 → #24
inhg2hpa	inHg → hPa	#24
Inverse cosine $\cos^{-1}$	ACOS	#1 → #29
Inverse sine $\sin^{-1}$	ASIN	#1 → #28
Inverse tangent $\tan^{-1}$	ATAN	#1 → #30
ISA temperature at X ft	ISA	#13
kg2lb	kg → lb	#1 → #22
km2nm	km → nm	#1 → #17

kmh2kt	kmh → kt	#1 → #19
kt2kmh	kt → kmh	#19
l2gal	L → gal	#1 → #23
LASTx — recall the X value used by the last operation	LASTx	#45
lb2kg	lb → kg	#22
m2ft	m → ft	#1 → #20
Mach: Y TAS kt, X OAT °C	MACH	#15
Magnetic = Y true − X variation (E+)	MAG	#1 → #12
mi2nm	mi → nm	#1 → #18
Multiplication ×	×	#44
nm2km	nm → km	#17
nm2mi	nm → mi	#18
ON — power on	ON	#8
Percent %	%	#55
Pressure alt: Y elevation, X altimeter inHg	PALT	#1 → #13
R↓ — roll the stack down	R↓	#38
Rate of descent: Y GS, X angle	ROD	#1 → #16
RCL — recall a memory / variable	RCL	#40
recip	RECIP	#12
Reciprocal x^{-1} (1/x)	1/x	#1 → #32
SETUP — angle unit, number format, display options	MODE	#56
SHIFT / 2nd — the next key uses its upper (shift-coloured) function	SHIFT	#1
Sine (uses the current angle unit)	SIN	#28

Slope % to degrees	SLOPE°	#27
Speed = Y dist ÷ X time	SPD	#3
Square root √	√x	#31
Square x^2	x^2	#1 → #31
Standard-rate bank angle	BANK	#1 → #26
Standard-rate turn radius (nm) at X kt	TURN R	#26
STO — store the value into a memory / variable (press the target next)	STO	#39
Subtract H.MMSS	HMS−	#1 → #6
Subtraction −	−	#52
Tangent (current angle unit)	TAN	#30
TAS: Y CAS, X altitude ft	TAS	#1 → #15
Time = Y dist ÷ X speed	TIME	#2
undo	UNDO	#46
UTC (Y) + offset (X)	TZ	#7
Wind triangle form	WIND	#9
x≷y — exchange X and Y	x ↔ y	#37
y^x — raise Y to the power X (RPN: base first, then exponent)	yx	#32
Δ% — percent change	Δ%	#1 → #55

Appendix C — Glossary

Ans
The result of the last calculation, reusable in the next.

AOS
Algebraic Operating System — operator precedence on a one-line display.

RPN
Reverse Polish Notation — operands first, then the operation; no = key.

Stack
The RPN registers X, Y, Z, T where numbers wait for operations.

TVM
Time Value of Money — n, i, PV, PMT, FV.

Natural display
Shows fractions, roots and powers as printed in textbooks.

Shift / 2nd
Prefix key that selects the function printed above a key.

Alpha
Prefix key that types the letter printed beside a key.

Radian
Angle unit; 2π rad = 360°.

Grad
Angle unit; 400 grad = 360°.

Engineering notation
Scientific notation with exponents that are multiples of 3.

Mantissa
The digits of a number in scientific notation.

Memory (M)

An independent register for running totals.

Mark-up

Increase over cost; margin is profit as a percent of selling price.

Grand total (GT)

Sum of all totals since the last clear.

nCr / nPr

Combinations / permutations of n items taken r at a time.

Factorial

n! = 1×2×…×n.

Reciprocal

1/x.

Standard deviation

Spread of data around the mean (σ population, s sample).

Key number

The small badge 1…N on each key in U-45 CALCULATORS, used throughout this manual.

Appendix D — Model notes & limits

Navigation & flight-computer RPN for training and planning exercises (not certified for real navigation). Formulas: haversine great-circle on a 3 440.065 nm sphere, standard wind-triangle trigonometry, ISA lapse 1.98 °C / 1 000 ft.

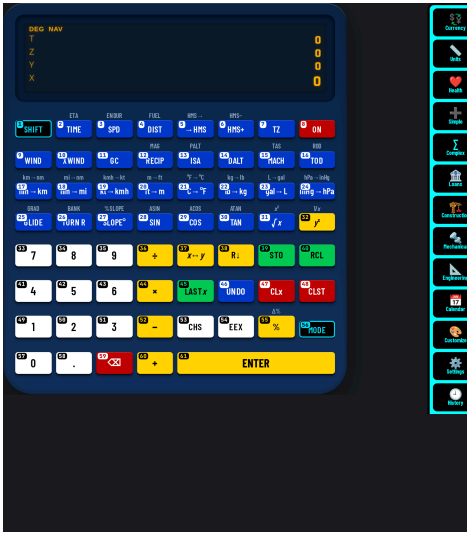
Times use the H.MMSS convention: 1.3000 means 1 h 30 min 00 s.

Planned (not yet working)

- **Stack history view**
- **Undo**
- **Equation library**
- **Speech read-out**
- **Colour-blind key theme**

U-45 CALCULATORS models are independent software recreations; they are not affiliated with any calculator manufacturer and carry no third-party trademarks.

Appendix E — Quick-reference card



#1 SHIFT	#2 TIME ⇧ ETA	#3 SPD ⇧ ENDUR
#4 DIST ⇧ FUEL	#5 →HMS ⇧ HMS→	#6 HMS+ ⇧ HMS−
#7 TZ	#8 ON	#9 WIND
#10 XWIND	#11 GC	#12 RECIP ⇧ MAG
#13 ISA ⇧ PALT	#14 DALT	#15 MACH ⇧ TAS
#16 TOD ⇧ ROD	#17 nm → km ⇧ km → nm	#18 nm → mi ⇧ mi → nm
#19 kt → kmh ⇧ kmh → kt	#20 ft → m ⇧ m → ft	#21 °C → °F ⇧ °F → °C
#22 lb → kg ⇧ kg → lb	#23 gal → L ⇧ L → gal	#24 inHg → hPa ⇧ hPa → inHg
#25 GLIDE ⇧ GRAD	#26 TURN R ⇧ BANK	#27 SLOPE° ⇧ %SLOPE
#28 SIN ⇧ ASIN	#29 COS ⇧ ACOS	#30 TAN ⇧ ATAN
#31 √x ⇧ x²	#32 yx ⇧ 1/x	#33 7
#34 8	#35 9	#36 ÷
#37 x ↔ y	#38 R↓	#39 STO
#40 RCL	#41 4	#42 5
#43 6	#44 ×	#45 LASTx
#46 UNDO	#47 CLx	#48 CLST
#49 1	#50 2	#51 3

#52 –	**#53** CHS	**#54** EEX
#55 % ⇧ Δ%	**#56** MODE	**#57** 0
#58 .	**#59** ⌫	**#60** +
#61 ENTER		

U-45 CALCULATORS · Calculator #37 · cut out and keep